

Algebra 1B

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1 Vector Spaces and Linear Maps

Definition. A **vector space** V over a field $\mathbb{F}$ is a set V with two operations, $+$ and $\times$.

All algebraic manipulation in a vector space ultimately comes from the Vector Space Axioms and Field Axioms.

1.1 Subspaces

Definition. Let V be a vector space over $\mathbb{F}$. A subset $U \subseteq V$ is a (linear) **subspace** if

- $0 \in U$
- U is closed under linear combinations, meaning

$$\forall u, w \in U, \lambda, \mu \in \mathbb{F} \quad \lambda u + \mu w \in U.$$

In this case, we write $U \leq V$.

Remark. Because addition and multiplication is defined, U is a vector space in its own right.

Any V has two special subspaces, V itself and the trivial subspace $\{0\}$. A subspace U with $U \neq V$ is said to be 'proper'.

1.2 Linear Maps

Definition. Let V, W be vector spaces over $\mathbb{F}$. A map $\phi : V \rightarrow W$ is a **linear map** if

$$\phi(\lambda v + \mu w) = \lambda \phi(v) + \mu \phi(w) \quad \lambda, \mu \in \mathbb{F}, v, w \in V.$$

If $V = W$, then ϕ is a linear operator on V .

Lemma.

- The identity map $\text{id}_V : V \rightarrow V$, defined $\text{id}_V(v) = v$, is a linear operator over V .
- If $\phi : V \rightarrow W$ is a linear map then

$$\phi(0) = 0, \quad \phi(-v) = -\phi(v).$$

- If $\phi : V \rightarrow W, \psi : W \rightarrow U$ are linear maps then $\psi \circ \phi : V \rightarrow U$ is also a linear map
- If $\phi : V \rightarrow W$ is a linear bijection then $\phi^{-1} : W \rightarrow V$ is also linear.

Definition. A linear bijection $\phi : V \rightarrow W$ is called a (linear) **isomorphism**.

We say V is **isomorphic to** W if an isomorphism $\phi : V \rightarrow W$ exists, written $V \cong W$.

1.3 Subspaces from Linear Maps

Definition. Let $\phi : V \rightarrow W$ be linear.

The **kernel** of ϕ is $\ker \phi \subseteq V$ given

$$\ker \phi = \{v \in V \mid \phi(v) = 0\}.$$

The **image** of ϕ is $\text{Im } \phi \subseteq W$ given by

$$\text{Im } \phi = \{\phi(v) \mid v \in V\}$$

Proposition. Let $\phi : V \rightarrow W$ be linear. Then

- $\ker \phi \leq V$
- $\text{Im } \phi \leq W$.

Proposition. A linear map $\phi : V \rightarrow W$ is injective if and only if $\ker \phi = \{0\}$

Corollary. A linear map $\phi : V \rightarrow W$ is an isomorphism if and only if $\ker \phi = \{0\}$ and $\text{Im } \phi = W$.

2 Bases, Coordinates and Matrices

2.1 On Lists

Definition. A **list** of vectors in a vector space V is a finite sequence $v_1, \dots, v_n$ of elements of V . A **sequence** is a map $\{1, \dots, n\} \rightarrow V$ defined $i \rightarrow v_i$

There are some differences between lists and (sub)sets:

- lists are ordered
- lists allow the same element to appear more than once

2.2 Spanning and Independent lists

Definition. Let $\alpha : v_1, \dots, v_n \in V$ be a list of vectors in a vector space V over $\mathbb{F}$.

The **span** of α is given by

$$\text{span } \alpha = \text{span}\{v_1, \dots, v_n\} := \left\{ \sum_{i=1}^n \lambda_i v_i \mid \lambda_1, \dots, \lambda_n \in \mathbb{F} \right\} \leq V.$$

More generally, for $S \subseteq V$, the span of S is

$$\text{span } S := \left\{ \sum_{i=1}^n \lambda_i s_i \mid s_1, \dots, s_n \in S, \lambda_1, \dots, \lambda_n \in \mathbb{F}, n \in \mathbb{N} \right\} \leq V.$$

$\text{span } S$ is the smallest subspace of V containing S :

$$\text{span } S = \bigcap_{u \leq V \mid S \subseteq u} u.$$

Definition. Let $\alpha : v_1, \dots, v_n \in V$ be a list of vectors in a vector space V over $\mathbb{F}$.

- α is **(linearly) independent** if, whenever $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ have

$$\sum_{i=1}^n \lambda_i v_i = 0 \text{ then } \lambda_1 = \dots = \lambda_n = 0.$$

and **(linearly) dependent** otherwise.

- α is **spanning** or **spans** if

$$\text{span } \alpha = V.$$

- α is a **basis** of V if it is both independent and spans.

Remark. Independence of a list $\alpha : v_1, \dots, v_n \in V$ says that no element $v_i \in \alpha$ can be expressed as another element multiplied by a scalar λ , or furthermore

$$\sum_{i=1}^n \lambda_i v_i \neq \sum_{i=1}^n \mu_i v_i.$$

unless $\lambda_i = \mu_i$ for all i , meaning there is a one-to-one mapping from the list $\lambda_1, \dots, \lambda_n$ to the span of α .

Lemma.

- A single vector v is independent if $v \neq 0$
- A two element list v_1, v_2 is dependent $\Leftrightarrow v_1 = \lambda v_2$ for some $\lambda \in \mathbb{F}$

2.3 Spanning and Independence in $\mathbb{F}^n$

Proposition. Let $\alpha : v_1, \dots, v_n \in \mathbb{F}^M$ and define $A \in M_{M \times N}(\mathbb{F})$ by $\text{col}_i(A) = v_i$. Let $\hat{A}$ be an REF matrix obtained from A by gaussian elimination. Then

- The list α is independent if and only if every column of $\hat{A}$ has a pivot.
- α is spanning if $\hat{A}$ has no zero rows.

2.4 Bases and Isomorphisms

Proposition. Let $\alpha : v_1, \dots, v_n$ be a list in V and define $\phi_\alpha : \mathbb{F}^n \rightarrow V$

$$\phi_\alpha(\mathbf{x}) \rightarrow \sum_{i=1}^n v_i x_i.$$

Then

- α is independent if and only if ϕ_α is injective
- α spans if and only if ϕ_α is surjective.

Hence α is a basis if and only if ϕ_α is an isomorphism.

Corollary. A list $\alpha : v_1, \dots, v_n \in V$ is a basis if and only if every $v \in V$ can be written as a linear combination

$$v = \sum_{i=1}^n x_i v_i$$

for unique $x_1, \dots, x_n \in \mathbb{F}$.

Proposition. Let $\phi : \mathbb{F}^n \rightarrow V$ be an isomorphism. Then $\phi = \phi_\alpha$ where $\alpha : \phi(e_1), \dots, \phi(e_n) \in V$ is a basis.

2.5 Bases and Coordinates

A basis allows us to represent abstract vectors by lists of scalars (coordinates).

Definition. Let $\alpha : v_1, \dots, v_n \in V$ be a basis of V . For $v \in V$, we call the vector $(x_1, \dots, x_n) \in \mathbb{F}^n$ defined by

$$v = \sum_{i=1}^n x_i v_i.$$

This is the **coordinate vector** of v with respect to α .

Remark.

- $v = \phi_\alpha(x_1, \dots, x_n)$ so

$$(x_1, \dots, x_n) = \phi_\alpha^{-1}(v).$$

- Reordering α gives a different basis with different coordinate vectors.

2.6 Bases, Linear Maps and Matrices

Definition. Let V, W be vector spaces over $\mathbb{F}$, $\alpha : v_1, \dots, v_n \in V$ be a basis of V , $\beta : w_1, \dots, w_m \in W$ be a basis of W , and $\phi : V \rightarrow W$ is a linear map. The **matrix** of ϕ with respect to α, β is $A \in M_{m,n}(\mathbb{F})$ given by

$$\phi(v_j) = \sum_{i=1}^m A_{ij} w_i.$$

This means the j^{th} column of A is $\phi(v_j)$ written in terms of β .

When $V = W$, we can take $\alpha = \beta$

Definition. Let V be a vector space over $\mathbb{F}$ with basis $\alpha = v_1, \dots, v_n$ and $\phi : V \rightarrow V$ is a linear map on V . The **matrix** of ϕ with respect to α is $A \in M_n(\mathbb{F})$ given by

$$\phi(v_j) = \sum_{i=1}^n A_{ij} v_i.$$

Proposition. Let $\alpha : v_1, \dots, v_n$ be a basis of V , $\beta : w_1, \dots, w_m$ be a basis of W , and $\phi : V \rightarrow W$ is a linear map with matrix A with respect to α, β .

- For $v \in V$ with coordinate vector x with respect to α , $\phi(v)$ has coordinates y with respect to β where

$$\underline{y} = A\underline{x}.$$

- We have

$$\phi = \phi_\beta \circ \phi_A \circ \phi_\alpha^{-1}$$

or, equivalently

$$\phi_A = \phi_\beta^{-1} \circ \phi \circ \phi_\alpha.$$

Proposition. Let V, W, U be vector spaces over $\mathbb{F}$ with bases α, β, γ respectively. Let $\phi : V \rightarrow W$ and $\psi : W \rightarrow U$ be linear maps with matrices A with respect to α, β and B with respect to β, γ .

Then $\psi \circ \phi$ has matrix BA with respect to α, γ .

2.7 Change of Basis

What happens if we want to compose two linear maps that use different bases?

Definition. Let $\alpha : v_1, \dots, v_n$ and $\alpha' : v'_1, \dots, v'_n$ be two bases of V . Then the change of basis matrix with respect to α, α' is $P \in M_n(\mathbb{F})$ such that

$$v'_j = \sum_{i=1}^n P_{ij} v_i.$$

This means the j^{th} column of P is v'_j written in terms of α .

Lemma. $\phi_P = \phi_{\alpha'}^{-1} \circ \phi_{\alpha}$ and so is an isomorphism, hence P is invertible.

Corollary. The change of basis matrix from α' to α is P^{-1} .

Proposition. Let P be a change of basis matrix from α to α' . Let $v \in V$ have coordinate vector x with respect to α and x' with respect to α' . Then

$$\underline{x} = P \underline{x'}.$$

Proposition. Let α, α' be bases of V with change of basis matrix P from α to α' . Let β, β' be bases of W with change of basis matrix Q from β to β' .

Let $\phi : V \rightarrow W$ is a linear map with matrix A with respect to α, β and A' with respect to α', β' . Then

$$A' = Q^{-1} A P.$$

Corollary. If α, α' are bases of V with change of basis matrix P from α to α' , then, if $\phi : V \rightarrow W$ is a linear operator on V with matrix A with respect to α and A' with respect to α' , then

$$A' = P^{-1} A P.$$

These results prompt the following definition.

Definition.

- Matrices $A, A' \in M_{M,N}(\mathbb{F})$ are **equivalent** if there exists invertible matrices $Q \in M_M(\mathbb{F}), P \in M_N(\mathbb{F})$ such that

$$A' = Q^{-1} A P.$$

- $A, A' \in M_N(\mathbb{F})$ are **similar** if there exists invertible $P \in M_N(\mathbb{F})$ such

that

$$A' = P^{-1}AP.$$

Both equivalence and similarity are equivalence relations.

2.8 Computations in $\mathbb{F}^n$

Example. When α is the standard bases of $\mathbb{F}^n$, and $\alpha' : v'_1, \dots, v'_n$ another basis, then P is easy to compute:

$$\text{col}_j(P) = v'_j.$$

The reason we care about all this: there may be a cunning choice of basis which gives a simpler matrix for a given linear map.

3 Dimension and Rank

3.1 Dimension

Lemma. (Fundamental Lemma)

Let $A \in M_{M,N}(\mathbb{F})$. If $n > m$ then there is a nonzero $x \in \mathbb{F}^n$ such that $Ax = 0$.

Theorem.

- If $v_1, \dots, v_n \in V$ are an independent list of vector, and $w_1, \dots, w_m$ are a spanning list, then $n \leq m$.
- If $v_1, \dots, v_n$ and $w_1, \dots, w_m$ are bases, then $n = m$.

Definition. Let V be a vector space.

- V is **finite dimensional** if it has a finite length basis and **infinite dimensional** otherwise
- If V is finite dimensional, the **dimension** of V , denoted $\dim V$ is the length of any basis of V .

Example.

- $\dim \mathbb{F}^n = n$
- $\dim \mathbb{F}^0 = \dim \{0\} = 0$
- $\dim \mathbb{F}[x] = \infty$

Dimension is preserved by isomorphism.

Proposition. Let V be a vector space over $\mathbb{F}$. Then $\dim V = n$ if and only if $\mathbb{F}^n \cong V$.

In this case, if $W \cong V$ then W is finite dimensional also with $\dim W = n$.

Proposition.

- If $v_1, \dots, v_n$ span V and $v_n \in \text{span}\{v_1, \dots, v_{n-1}\}$ then $v_1, \dots, v_{n-1}$ span V .
- If $v_1, \dots, v_{n-1}$ is an independent list and $v_n \notin \text{span}\{v_1, \dots, v_{n-1}\}$ then $v_1, \dots, v_n$ is also independent.

Definition. Let α be a list of vectors in V .

- a **sublist** of α is a list obtained by deleting some elements of α
- a list β **extends** α if β is obtained by appending vectors to α .

Theorem. Let β be a finite spanning list of vectors in a vector space V extending a (possibly empty) independent list α . Then there is a basis γ of V which

- extends α
- is a sublist of β .

Corollary.

- Any finite spanning list contains a basis
- If V is finite dimensional, any independent list can be extended to a basis.

Corollary. Let $\dim V = n$.

- Any spanning list of length n is a basis.
- Any independent list of length n is a basis.

Theorem. Let V be finite dimensional and $W \leq V$. Then W is finite dimensional also and $\dim W \leq \dim V$.

Furthermore, $\dim W = \dim V \Leftrightarrow W = V$.

Remark. If two vector spaces have the same dimension, they are only equal if one is a subspace of the other. Equal dimension alone is not enough to show they are equal.

Proposition. Let $\alpha : v_1, \dots, v_n \in \mathbb{F}^m$ and define $A \in M_{m,n}(\mathbb{F})$ by $\text{col}_j(A) = \bar{v}_j$.

Let $\hat{A}$ be a REF matrix obtained from A by gaussian elimination with r pivots in cols $j_1 < j_2 < \dots < j_r$. Then $v_{j_1}, \dots, v_{j_r}$ is a basis for $\text{span } \alpha \leq \mathbb{F}^M$.

3.2 Rank

Definition. The **rank** of a linear map $\phi : V \rightarrow W$ is

$$\text{rank } \phi := \dim \text{Im } \phi.$$

Lemma. Given $\phi : V \rightarrow W$ and isomorphisms $f : V' \rightarrow V$ and $g : W' \rightarrow W$, set $\phi' = g^{-1} \circ \phi \circ f : V' \rightarrow W'$. Then

$$\text{rank } \phi = \text{rank } \phi'.$$

Definition. Let $A \in M_{M,N}(\mathbb{F})$.

- The **column space** of A is the span of the columns of A .

$$\text{colspace } A := \left\{ \sum_{j=1}^n x_j \text{col}_j(A) \mid x_1, \dots, x_n \in \mathbb{F} \right\}.$$

The **column rank** of A is $\text{colrank } A := \dim \text{colspace } A$.

- The **row space** of A is the span of the rows of A

$$\text{rowspace } A := \left\{ \sum_{j=1}^m x_j \text{row}_j(A) \mid x_1, \dots, x_n \in \mathbb{F} \right\}.$$

The **row rank** of A is $\text{row rank } A := \dim \text{rowspace } A$.

- $\text{rank } A$ is the number of pivots in the REF of A .

Theorem. Let $A \in M_{m,n}(\mathbb{F})$. Then $\text{rank } A$ is equal to:

- $\text{rank } \phi_A$
- $\text{colrank } A$
- $\text{rowrank } A$
- The smallest number r such that

$$A = BC$$

with $B \in M_{M,r}, C \in M_{r,N}$.

Corollary. For $A \in M_{M,N}$, $\text{rank } A = \text{rank } A^T$.

Remark. To find the rank of a matrix A , and the 'smallest' matrices B and C such that $A = BC$, reduce A into RREF. Then the number of pivots is $r = \text{rank } A$. The columns of B are the columns of A where there is a pivot in $\hat{A}$, and the columns of C are the first r rows of $\hat{A}$.

Proposition.

- Let $\phi : V \rightarrow W$ be a linear map with matrix A with respect to some bases α, β . Then

$$\text{rank } A = \text{rank } \phi.$$

- Let $A, B \in M_{M,N}(\mathbb{F})$ be equivalent. Then

$$\text{rank } A = \text{rank } B.$$

Remark. The above statement has a converse: If $\text{rank } A = r$ then A is equivalent to $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$. Additionally, matrices of the same size and rank are equivalent under an equivalence relation.

3.3 The Rank-Nullity Theorem

Definition. The **nullity** of a linear map ϕ is

$$\text{null } \phi = \dim \ker \phi.$$

Theorem. (Rank-Nullity Theorem)

If V is finite dimensional and $\phi : V \rightarrow W$ is linear, then

$$\text{rank } \phi + \text{null } \phi = \dim V.$$

Remark. The dimension of W need not be finite.

Theorem. Let $\phi : V \rightarrow V$ be a linear operator on finite dimensional V . Then the following are equivalent:

- ϕ is an isomorphism
- ϕ is surjective
- ϕ is injective
- $\ker \phi = \{0\}$

Remark. This result fails if V is infinite dimensional.

3.4 Applications to Sums and Products

Definition. Let V, W be vector spaces over $\mathbb{F}$. Then the **product** of V and W is $V \times W$ with addition and scalar multiplication defined componentwise

$$(v, w) + (v', w') = (v + v', w + w')$$

$$\lambda(v, w) = (\lambda v, \lambda w).$$

$V \times W$ is a vector space over $\mathbb{F}$ with this addition and scalar multiplication.

Proposition. If V, W are finite dimensional, so is $V \times W$ and

$$\dim(V \times W) = \dim V + \dim W.$$

Definition. Let $U, W \leq V$. Then the **sum** of U and W is

$$U + W := \{u + w | u \in U, w \in W\}.$$

Remark. $U + W$ is itself a subspace of V .

Proposition. (Dimensional Formula)

If $U, W \leq V$ are finite dimensional. Then

$$\dim(U + W) + \dim U \cap W = \dim U + \dim W.$$

4 Determinants and Inverses

4.1 Determinants: the Sum Formula

Definition. Let $A = (a_{ij}) \in M_N(\mathbb{F})$. The determinant of A , denoted $\det A$ or $|A|$ is

$$\det A = \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{\sigma(1)1} \dots a_{\sigma(n)n}.$$

This is a signed product of one entry from each column, in a different row each time, all added together.

Remark. There are $n!$ summands so this formula is not easy to use as n gets larger. Fortunately, there are other ways to compute $\det A$.

Proposition. If $A = \begin{pmatrix} A' & * \\ 0 & A'' \end{pmatrix}$ with $A' \in M_k(\mathbb{F})$, $A'' \in M_{n-k}(\mathbb{F})$ is block diagonal (or upper triangular) then $\det A = \det A' \det A''$.

Proposition. $\det A^T = \det A$.

4.2 Properties of Determinants

We view $\det A$ as a function of the columns of A :

write $A = (\underline{c}_1, \dots, \underline{c}_n)$ $\underline{c}_j = \text{col}_j(A)$ and view $\det : (\underline{c}_1, \dots, \underline{c}_n) \mapsto \det A$

Proposition. $\det A = \det(\underline{c}_1, \dots, \underline{c}_n)$ is

- **multilinear** (linear in each slot separately): if $\underline{c}_j = \lambda \underline{v} + \mu \underline{w}$ where $\underline{v}, \underline{w}$ are column vectors, $\lambda, \mu \in \mathbb{F}$, then $\forall j, 1 \leq j \leq n$

$$\det(\underline{c}_1, \dots, \underline{c}_j, \dots, \underline{c}_n) = \lambda \det(\underline{c}_1, \dots, \underline{v}, \dots, \underline{c}_n) + \mu \det(\underline{c}_1, \dots, \underline{w}, \dots, \underline{c}_n)$$

- **alternating**: if $\underline{c}_i = \underline{c}_j$ for some $i \neq j$, then

$$\det(\underline{c}_1, \dots, \underline{c}_n) = 0.$$

Corollary. Considering $A \in M_N(\mathbb{F})$ as a list of row vectors $\underline{r}_1, \dots, \underline{r}_n \in M_N(\mathbb{F})$, and the function $\det A = \det(\underline{r}_1, \dots, \underline{r}_n)$ is

- **multilinear** (linear in each slot separately): if $\underline{r}_j = \lambda \underline{v} + \mu \underline{w}$ where $\underline{v}, \underline{w}$ are column vectors, $\lambda, \mu \in \mathbb{F}$, then $\forall j, 1 \leq j \leq n$

$$\det(\underline{r}_1, \dots, \underline{r}_j, \dots, \underline{r}_n) = \lambda \det(\underline{r}_1, \dots, \underline{v}, \dots, \underline{r}_n) + \mu \det(\underline{r}_1, \dots, \underline{w}, \dots, \underline{r}_n)$$

- **alternating**: if $\underline{r}_i = \underline{r}_j$ for some $i \neq j$, then

$$\det(\underline{r}_1, \dots, \underline{r}_n) = 0.$$

Proposition. If $\hat{A}$ is obtained from A by

- $\hat{\underline{c}}_i = \lambda \underline{c}_i$ then $\det \hat{A} = \lambda \det A$
- $\hat{\underline{c}}_i = \underline{c}_i + \lambda \underline{c}_j$ then $\det \hat{A} = \det A$
- $\underline{c}_i \leftrightarrow \underline{c}_j, (i \neq j)$ (swap columns i and j) then $\det \hat{A} = -\det A$

Remark. Since $\det A^T = \det A$, this works just as well for EROs as it does for ECOs. This gives us a method to compute the determinant:

Use gaussian elimination (also using ECOs) to turn A into an upper or lower triangular $\hat{A}$, keeping track of the changes to the determinant.

Corollary. If $\hat{A}$ is obtained from A by a mix of EROs and ECOs. Then

$$\det \hat{A} = 0 \Leftrightarrow \det A = 0.$$

Corollary. If $\hat{A}$ is obtained from A by permuting the columns of A (or the rows) by $\sigma \in S_n$ then

$$\det \hat{A} = \text{sign } \sigma \det A.$$

4.3 Characterisation of det

Fix $n > 0$, and let $I := I_n \in M_n(\mathbb{F})$, the identity matrix.

Theorem. Let $D : M_n(\mathbb{F}) \rightarrow \mathbb{F}$ be a function that is multilinear and alternating in columns. Then

$$D(A) = D(I) \det A.$$

Corollary. If $D : M_n(\mathbb{F}) \rightarrow \mathbb{F}$ is multilinear and alternating and $D(I) = 1$, then

$$D = \det .$$

Theorem. (Product formula)
If $A, B \in M_n(\mathbb{F})$, then

$$\det(AB) = \det A \det B.$$

Corollary.

- If $P \in M_m(\mathbb{F})$ is invertible, then $\det(P^{-1}) = (\det P)^{-1}$.
- If A and B are similar, then $\det A = \det B$.

Remark. If $\phi : V \rightarrow V$ is a linear operator and A, B are the matrices of ϕ with respect to two bases, then A and B are similar, so $\det A = \det B$.

Hence the following is well defined.

Definition. Let $\phi : V \rightarrow V$ be a linear map, and V be finite dimensional. The **determinant** of ϕ , $\det \phi$, is

$$\det \phi := \det A$$

where A is the matrix of ϕ with respect to some basis.

4.4 Minors, Adjugates and Inverses

We introduce a method for computing the determinant, and the inverse of a matrix.

Definition. For $A = (a_{ij}) \in M_n(\mathbb{F})$ the (i, j) -minor is

$$\mu_{ij}(A) := \det M_{ij}(A)$$

where $M_{ij}(A)$ is the matrix obtained by deleting the i^{th} row and j^{th} column of A .

Proposition. (Column and Row Expansion of Determinant)

- For any fixed column j

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \mu_{ij}(A).$$

- For any fixed row i

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \mu_{ij}(A).$$

Definition. The **adjugate** of a matrix $A \in M_N(\mathbb{F})$ is

$$\text{adj}(A) = (b_{ij}) \in M_N(\mathbb{F})$$

where

$$b_{ij} = (-1)^{i+j} \mu_{ji}(A)$$

We say $(-1)^{i+j} \mu_{ji}(A)$ is the (i, j) -cofactor and so $\text{adj}(A)$ is the transpose of the cofactor matrix.

Theorem.

$$\text{adj}(A)A = \det(A)I.$$

Corollary. If $\det A \neq 0$ and then A is invertible and

$$A^{-1} = \frac{1}{\det A} \text{adj}(A).$$

Remark. By the product formula, if A is invertible then $\det A \neq 0$. We also have the converse:

$$A \text{ is invertible} \Leftrightarrow \det A \neq 0.$$

5 Eigenvalues and Diagonalisation

5.1 Fundamental Theorem of Algebra

Theorem. (Fundamental Theorem of Algebra)

Let $f(x) \in \mathbb{C}[x]$ be a non-constant polynomial of degree $d > 0$. The following equivalent statements hold:

- $f(x)$ has at least one complex root
- $f(x)$ has precisely d roots counted with multiplicity
- $f(x)$ factorizes as a product of linear terms

5.2 Eigenvalues and Characteristic Polynomials of Square Matrices

Definition. Let $A \in M_n(\mathbb{F})$. A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of A if there exists a non-zero vector v such that

$$Av = \lambda v.$$

In this case the vector v is an **eigenvector** of A with **eigenvalue** λ or a **λ -eigenvector**.

Remark. If v is an eigenvector with eigenvalue λ then every nonzero scalar multiple av of v is also a λ -eigenvector.

There is a method for computing eigenvectors!

Definition. The characteristic polynomial of a matrix $A \in M_n(\mathbb{F})$ is

$$\Delta_A(t) = \det(A - tI) \in \mathbb{F}[t].$$

Where $I \in M_n(\mathbb{F})$ is the identity matrix $I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$

Proposition. The eigenvalues of A are the roots of $\Delta_A(t)$ in $\mathbb{F}$.

Lemma. If $A \in M_n(\mathbb{F})$ then $\Delta_A(t)$ is a degree n polynomial, with leading coefficient $(-1)^n$. Therefore $\Delta_A(t)$ has at most n roots. If $\mathbb{F} = \mathbb{C}$ then $\Delta_A(t)$ has precisely n roots counted with multiplicity.

Lemma. Similar matrices have the same characteristic polynomial, hence the same eigenvalues.

Definition. Let V be a vector space over $\mathbb{F}$, and $\phi : V \rightarrow V$ be a linear map.

- $\lambda \in \mathbb{F}$ is an **eigenvalue** of ϕ if there exists nonzero $v \in V$ such that

$$\phi(v) = \lambda v.$$

Then v is a λ -**eigenvector** of ϕ .

- The λ -eigenspace of ϕ is

$$\begin{aligned} E_\phi(\lambda) &= \{v \in V \mid \phi(v) = \lambda v\} \\ &= \ker(\phi - \lambda \text{id}_V). \end{aligned}$$

Lemma. Let $A \in M_n(\mathbb{F})$, V be a vector space over $\mathbb{F}$, $\phi : V \rightarrow V$ be a linear map. If A is the matrix of ϕ in some basis α then A, ϕ have the same eigenvalues and eigenvectors.

Definition. The **characteristic polynomial** of $\phi : V \rightarrow V$ is

$$\Delta_\phi(t) = \det(\phi - t \text{id}_V).$$

Equivalently, if A represents ϕ in some basis $\Delta_\phi(t) := \Delta_A(t)$.

Remark. The characteristic polynomial ϕ is well defined because similar matrices have equal determinants.

5.3 Diagonalisation

Let $\phi : V \rightarrow V$ be a linear map and $A \in M_n(\mathbb{F})$, the matrix of ϕ with respect to some basis $v_1, \dots, v_n$. Then the following are equivalent.

- v_i is a λ_i -eigenvector for $i = 1, \dots, n$.

- A is diagonal and $A = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}$

Definition.

- A linear operator $\phi : V \rightarrow V$ is **diagonalisable** if there is a basis consisting of eigenvectors of ϕ . In this case we call it an **eigenbasis**.
- $A \in M_n(\mathbb{F})$ is **diagonalisable** (over $\mathbb{F}$) if there is an invertible $P \in$

$M_n(\mathbb{F})$ such that

$$P^{-1}AP \text{ is diagonal.}$$

We say that A is **diagonalised** by P .

Proposition. If $\phi : V \rightarrow V$ has a matrix A with respect to a basis β . Then the following are equivalent:

- ϕ is diagonalisable
- ϕ_A is diagonalisable
- A is diagonalisable

Remark. Many matrices are not diagonalisable over $\mathbb{R}$, but are diagonalisable over $\mathbb{C}$. This happens when the characteristic polynomial has no real roots.

5.4 Applications of Diagonalisation

Diagonalisation can be extremely helpful for efficient calculation of powers of matrices.

Example.

Suppose A is diagonalisable, with $P^{-1}AP = D$ being diagonal. Furthermore P has columns j equal to the λ_j eigenvectors of A . Then $A = PDP^{-1}$, so for $k > 0$

$$\begin{aligned} A^k &= (PDP^{-1})(PDP^{-1})\dots(PDP^{-1}) \\ &= PD^kP^{-1} \\ \text{with } D^k &= \begin{pmatrix} \lambda_1^k & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n^k \end{pmatrix}. \end{aligned}$$

This allows us to calculate A^k explicitly with only 2 matrix multiplications.

5.5 Multiplicity of Eigenvalues and Diagonalisation

We want a necessary and sufficient condition for a matrix or operator to be diagonalisable.

If $\dim V = n$ and $\phi : V \rightarrow V$ is linear, then an eigenbasis needs n linearly independent eigenvectors.

Theorem. If $v_1, \dots, v_n$ are eigenvectors ϕ with distinct eigenvalues $\lambda_1, \dots, \lambda_n$ then $v_1, \dots, v_n$ are independent.

Remark. This can be generalized: if we combine independent list from each eigenspace we get an independent list.

Corollary. Let $n = \dim V$ and $\phi : V \rightarrow V$ be a linear operator. If

$$\Delta_\phi(t) = (t - \lambda_1) \dots (t - \lambda_n)$$

with λ_i distinct for all i , then ϕ is diagonalisable.

Definition. Let λ be an eigenvector of $\phi : V \rightarrow V$. Then

- the **algebraic multiplicity** of λ denoted $am(\lambda)$ is the largest k such that $(\lambda - t)^k$ divides $\Delta_\phi(t)$.
- the **geometric multiplicity** of λ is $gm(\lambda) := \dim E_\phi(\lambda)$.

Remark.

- $\dim E_\phi(\lambda)$ is the maximal number of independent λ -eigenvectors.
- Suppose λ is a root of $\Delta_\phi(t)$. Then $(t - \lambda)$ divides $\Delta_\phi(t)$ so

$$am(\lambda) \geq 1.$$

If $E_\phi(\lambda) \neq 0$ then

$$gm(\lambda) \geq 1.$$

Proposition. For any $\lambda \in \mathbb{F}$ and any linear map ϕ :

$$am(\lambda) \geq gm(\lambda).$$

Theorem. Let $\phi : V \rightarrow V$ be a linear operator. Then ϕ is diagonalisable if and only if $\Delta_\phi(t)$ is a product of linear factors and $am(\lambda) = gm(\lambda)$ for all λ .

6 Inner Product Spaces

6.1 Inner Products

Recall that the dot (or scalar) product on $\mathbb{R}^n$ is the map

$$\begin{aligned} \mathbb{R}^n \times \mathbb{R}^n &\rightarrow \mathbb{R} \\ x \cdot y &= \sum_{i=1}^n x_i y_i. \end{aligned}$$

Then we can define the length of x to be $|x| = \sqrt{x \cdot x}$, and the angle between two vectors with $x \cdot y = |x||y| \cos \theta$.

We also have the dot product on $\mathbb{C}^n$, defined as

$$x \cdot y = \sum_{i=1}^n \bar{x}_i y_i.$$

Note that $x \cdot x = \sum_{i=1}^n \bar{x}_i x_i = \sum_{i=1}^n |x_i|^2$.

Definition. Let V be a vector space over $\mathbb{F}$ which we assume to be either $\mathbb{R}$ or $\mathbb{C}$. An **inner product** on V is a map $V \times V \rightarrow \mathbb{F} : (v, w) \mapsto \langle v, w \rangle$ which is

- **conjugate symmetric:** $\langle w, v \rangle = \overline{\langle v, w \rangle}$
- **linear in the second argument:**

$$\langle v, \lambda u + \mu w \rangle = \lambda \langle v, u \rangle + \mu \langle v, w \rangle$$

- **positive definite:** For all $v \in V$, $\langle v, v \rangle \geq 0$ with equality if and only if $v = 0$

A vector space with a fixed inner product is called an **inner product space**.

An inner product space comes with an inner product already defined, denoted $\langle \cdot, \cdot \rangle$.

Remark. $\langle v, v \rangle = \overline{\langle v, v \rangle}$ so the inner product of a vector on itself is always real.

- Suppose $\mathbb{F} = \mathbb{R}$. Then conjugate symmetry is just symmetry, and so $\langle v, w \rangle = \langle w, v \rangle$. Furthermore this means the inner product is linear in both arguments.
- Now suppose $\mathbb{F} = \mathbb{C}$. Then

$$\langle \lambda u + \mu v, w \rangle = \bar{\lambda} \langle u, w \rangle + \bar{\mu} \langle v, w \rangle.$$

Definition. Let v be an inner product space over $\mathbb{R}$.

- The **norm** of a vector v is $\|v\| = \sqrt{\langle v, v \rangle} \geq 0$.
If $\|v\| = 1$ then we say v is **unit-length**, and $\frac{v}{\|v\|}$ is the **normalisation** of v .
- Let $v, w \in V$. If $\langle v, w \rangle = 0$, then v is **orthogonal** to w , written $v \perp w$.

6.2 Cauchy-Schwarz Inequality

Theorem. Let V be an inner product space. For $v, w \in V$,

$$|\langle v, w \rangle| \leq \|v\| \|w\|$$

with equality if and only if v and w are dependent.

Corollary. (Triangle Inequality) For $v, w \in V$ we have

$$\|v + w\| \leq \|v\| + \|w\|$$

with equality if and only if v and w are dependent.

6.3 Orthogonality

Definition. A list $u_1, \dots, u_k$ in an inner product space V is **orthonormal** if for $1 \leq i, j \leq k$

$$\langle u_i, u_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Remark. A list is orthonormal if every u_i has unit length and distinct u_i, u_j are orthogonal.

Lemma. For an orthonormal basis $u_1, \dots, u_n$ in an inner product space V , for a vector $v \in V$ we have

$$v = \sum_{i=1}^n \langle u_i, v \rangle u_i.$$

Lemma. Any orthonormal list is linearly independent.

Proposition. If $u_1, \dots, u_n$ is an orthonormal basis for V and

$$v = \sum_{i=1}^n x_i u_i$$

$$w = \sum_{i=1}^n y_i u_i$$

then

$$\langle v, w \rangle = \sum_{i=1}^n \bar{x}_i y_i = x \cdot y$$

Theorem. (Gram-Schmidt Orthogonalisation) Let $v_1, \dots, v_n$ be an independent list in an inner product space V . Then there exists an orthonormal

list $u_1, \dots, u_n$ such that for all $1 \leq k \leq n$ we have

$$\text{span} \{v_1, \dots, v_k\} = \text{span} \{u_1, \dots, u_k\}.$$

We construct $u_1, \dots, u_k$ inductively:

- For $k = 1$ we set $w_1 = v_1$, and $u_1 = \frac{w_1}{\|w_1\|}$
- For $k > 1$, suppose we have defined $u_1, \dots, u_{k-1}$ already and set

$$\begin{aligned} w_k &= v_k - \sum_{j=1}^{k-1} \langle u_j, v_k \rangle u_j \\ &= v_k - \sum_{j=1}^{k-1} \frac{\langle w_j, v_k \rangle}{\|w_j\|^2} w_j. \end{aligned}$$

$$\text{and } u_k = \frac{w_k}{\|w_k\|}.$$

Corollary. Any finite dimensional inner product space has an orthonormal basis.

Remark. In practise it is easier to start with $v_1, \dots, v_n$, then produce an $w_1, \dots, w_n$ satisfying $w_i \perp w_j (i \neq j)$ then finally set $u_i = \frac{w_i}{\|w_i\|}$ for every i to get an orthonormal list.

Definition. A matrix $Q \in M_n(\mathbb{R})$ is orthogonal if $Q^T Q = \mathbb{I}_n$

Remark. Q is orthogonal if and only if the columns of Q are orthonormal with respect to the dot product.

Theorem. (QR Decomposition)

Let $A \in M_n(\mathbb{R})$ be invertible. Then there exists orthogonal Q and upper-triangular R such that

$$A = QR.$$

Remark. To compute Q and R in practise it is simplest to first find Q , then at $Q^T = Q^{-1}$, we have

$$R = Q^T A.$$

6.4 Orthogonal Complements

Let V be an inner product space and let $U \leq V$. Recall $u, v \in V$ are orthogonal, $u \perp v$, if $\langle u, v \rangle = 0$.

Definition. The **orthogonal complement** $U^\perp$ of u in V is

$$U^\perp = \{v \in V \mid \langle u, v \rangle = 0 \text{ for } u \in U\}.$$

Proposition.

- $U^\perp$ is a subspace of V
- $U \cap U^\perp = \{0\}$
- $U \subseteq (U^\perp)^\perp$

Theorem. If $U \leq v$ is a finite dimensional subspace then $V = U + U^\perp$.

Corollary. Let V be a finite dimensional inner product space.

- $\dim U^\perp = \dim V - \dim U$
- $U = (U^\perp)^\perp$

7 Linear Operators on Inner Product Spaces

Continue to assume that $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

7.1 Adjoints

Let V be an inner product space.

Notation. $L(V) := \{f : V \rightarrow V \mid f \text{ linear}\}.$

Note that $L(V)$ is itself a vector space.

Lemma. (Nondegeneracy Lemma)

For $v \in V$ the following are equivalent.

- $\langle v, V \rangle = \{0\}$
- $v = 0$.

Definition. Let $\phi \in L(V)$. An **adjoint** to ϕ is an operator $\phi^* \in L(V)$ such that for $v, w \in V$

$$\langle \phi^*(v), u \rangle = \langle u, \phi(w) \rangle.$$

Equivalently,

$$\langle w, \phi^*(v) \rangle = \langle \phi(w), v \rangle.$$

Proposition. Let $\dim V < \infty$ and $\phi \in L(V)$. Then

- ϕ has a unique adjoint ϕ^*
- Let $\alpha : u_1, \dots, u_n$ be an orthonormal basis of V with respect to which ϕ has matrix $A = (a_{ij})$. Then ϕ^* has matrix

$$A^\dagger := \bar{A}^T.$$

Corollary. Let $A \in M_n(\mathbb{F})$. Then

$$\begin{aligned} (\phi_A)^* &= \phi_{\bar{A}^T} \quad \text{for } \mathbb{F} = \mathbb{C} \\ (\phi_A)^* &= \phi_{A^T} \quad \text{for } \mathbb{F} = \mathbb{R}. \end{aligned}$$

Proposition. Let $\phi, \psi \in L(V)$. Then

- $(\phi \circ \psi)^* = \psi^* \circ \phi^*$
- $(\phi + \lambda\psi)^* = \phi^* + \bar{\lambda}\psi^*$
- $(\phi^*)^* = \phi$
- $\text{id}_V^* = \text{id}_V$

Definition. Let V be an inner product space.

- Let $\phi \in L(V)$.
 - ϕ is **self-adjoint** if $\phi = \phi^*$
 - ϕ is **skew-adjoint** if $\phi = -\phi^*$
- Let $A \in M_n(\mathbb{F})$. If $\mathbb{F} = \mathbb{C}$

- We say A is **Hermitian** if

$$A = A^\dagger = \bar{A}^T.$$

- We say A is **skew-Hermitian** if

$$A = -A^\dagger = -\bar{A}^T.$$

If $\mathbb{F} = \mathbb{R}$:

- We say A is **symmetric** if

$$A = A^T.$$

- We say A is **skew-symmetric** if

$$A = -A^T.$$

Remark. For $\phi \in L(V)$ and $A \in M_n(\mathbb{F})$

ϕ is self-adjoint $\Leftrightarrow A$ is Hermitian or symmetric.

Similarly,

ϕ is skew-adjoint $\Leftrightarrow A$ is skew-Hermitian or skew-symmetric.

Example.

- $\text{id}_V^* = \text{id}_V$ so id_V is self adjoint
- $\phi \circ \phi^*$ is self-adjoint

$$(\phi \circ \phi^*)^* = (\phi^*)^* \circ \phi^* = \phi \circ \phi^*.$$

- for $\mathbb{F} = \mathbb{C}$

ϕ is self-adjoint $\Leftrightarrow i\phi$ is skew-adjoint.

Definition.

- We say $\phi \in L(V)$ is an **orthogonal transformation** if $\phi^* = \phi^{-1}$ and $\mathbb{F} = \mathbb{R}$
- We say $\phi \in L(V)$ is an **unitary transformation** if $\phi^* = \phi^{-1}$ and $\mathbb{F} = \mathbb{C}$
- We say $A \in M_n(\mathbb{F})$ is **orthogonal** if

$$A^T A = \text{id} \quad \text{and} \quad \mathbb{F} = \mathbb{R}.$$

- We say $A \in M_n(\mathbb{F})$ is **unitary** if

$$\bar{A}^T A = \text{id} \quad \text{and} \quad \mathbb{F} = \mathbb{C}.$$

Remark. Let ϕ have matrix A with respect to an orthonormal basis. ϕ is orthogonal or unitary iff A is orthogonal or unitary.

7.2 The Spectral Theorem

Definition. Let $\phi \in L(V)$. Then $U \leq V$ is **ϕ -invariant** if $\phi(U) \leq U$. In other words $\phi(u) \in U$ for $u \in U$.

Lemma. If $\phi, \psi \in L(V)$ and

$$\phi \circ \psi = \psi \circ \phi$$

then $E_\phi(\lambda)$ is ψ -invariant for all $\lambda \in \mathbb{F}$.

Lemma. Let V be finite dimensional and $\phi \in L(V)$. If $U \leq V$ is ϕ -invariant then $U^\perp$ is ϕ^* -invariant.

Definition. Let V be a finite dimensional inner product space. A linear operator ϕ is **normal** if it commutes with its adjoint:

$$\phi \circ \phi^* = \phi^* \circ \phi.$$

Example.

- Self adjoint and skew-adjoint operators are normal
- Unitary and orthogonal transformations are normal.

Proposition. Let V be finite dimensional and $\phi \in L(V)$. If ϕ is normal then $E_\phi(\lambda)^\perp$ is ϕ -invariant.

Definition. Let V be finite dimensional. A linear operator $\phi \in L(V)$ is **orthogonally diagonalisable** if V admits an orthonormal basis of eigenvectors for ϕ .

Proposition. Let V be finite dimensional over $\mathbb{F}$. Let $\phi \in L(V)$ be orthogonally diagonalisable. Then ϕ is normal, and if $\mathbb{F} = \mathbb{R}$ then ϕ is self-adjoint.

Theorem. (Spectral Theorem over $\mathbb{C}$)

Let V be a finite-dimensional inner product space of dimension n over $\mathbb{C}$, and let $\phi \in L(V)$.

- ϕ is orthogonally diagonalisable if and only if ϕ is normal
- If $A \in M_n(\mathbb{C})$ is Hermitian then there is a unitary matrix P where $P^{-1}AP$ which is diagonal.

Now, to prove the Spectral Theorem over $\mathbb{R}$ we need a way to guarantee a matrix has an eigenvector.

Lemma. Let V be a complex inner product space and let $\phi \in L(V)$ be self-adjoint.

- Every eigenvalue of ϕ is real
- If u, v are eigenvectors of eigenvalues λ, μ with $\lambda \neq \mu$ then $u \perp v$.

Proposition. A self-adjoint operator on a real finite dimensional inner product space V has a real eigenvalue.

Remark. The requirements for ϕ to be diagonalisable over $\mathbb{R}$ are stronger because we need stronger requirements to guarantee an eigenvalue exists.

Theorem. (Spectral Theorem over $\mathbb{R}$)

Let V be a finite-dimensional inner product space of dimension n over $\mathbb{F}$, and let $\phi \in L(V)$.

- ϕ is orthogonally diagonalisable if and only if ϕ is self-adjoint.
- If $A \in M_n(\mathbb{R})$ is symmetric then there is an orthogonal matrix P where $P^{-1}AP$ is diagonal.